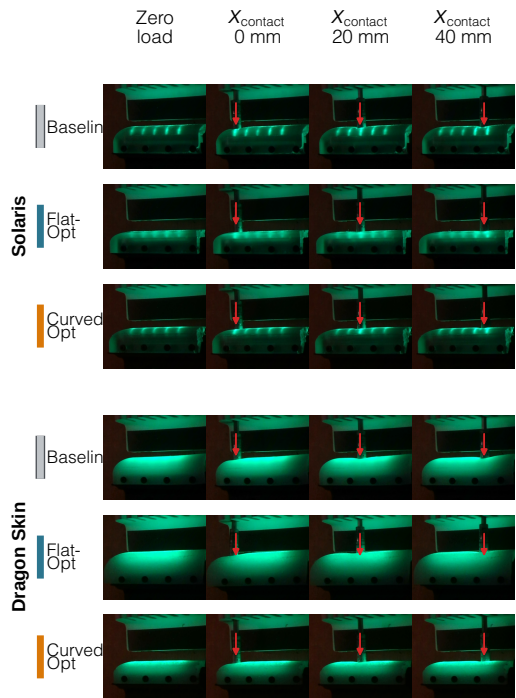
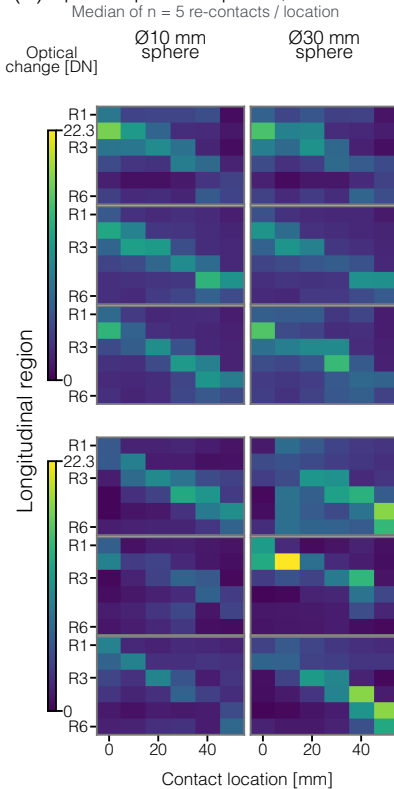


(a) Optical signatures across contact locations



(b) Spatial optical response, 2–15 N



(c) Contact-location decoding

$n = 30$ held-out predictions / matrix (5 / location)

